Tutorials: DEEP LEARNING

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1 Single Neuron and Basic Forward Pass

Exercise 1: Output of a Single Neuron

Suppose a neuron computes

$$z = w_1 x_1 + w_2 x_2 + b, \qquad y = \sigma(z),$$

where σ is the sigmoid activation

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

Given:

$$x_1 = 2, \quad x_2 = -1, \quad w_1 = 0.3, \quad w_2 = -0.5, \quad b = 0.1.$$

Tasks:

1. Compute the weighted sum z.
2. Compute the output $y = \sigma(z)$.

2 Multi-Neuron Layer Forward Pass

Exercise 2: Two-Neuron Layer with ReLU

We have a layer with two neurons:

$$z_1 = 1.0x_1 + 2.0x_2 - 1.0, \qquad z_2 = -2.0x_1 + 0.5x_2 + 0.2.$$

Input:

$$x_1 = 1, \quad x_2 = 3.$$

Activation: $\mathrm{ReLU}(z) = \max(0, z)$.
Tasks:

1. Compute z_1 and $a_1 = \mathrm{ReLU}(z_1)$.
2. Compute z_2 and $a_2 = \mathrm{ReLU}(z_2)$.
3. Write the vector output $a = (a_1, a_2)$.

3 Loss Function and Gradients

Exercise 3: Mean Squared Error Loss

Given target $y = 5$ and prediction $\hat{y} = 3.4$, with loss:

$$L = \frac{1}{2}(y - \hat{y})^2.$$

Task: Compute L.

Exercise 4: Derivative of MSE Loss

Given:

$$L = \frac{1}{2}(y - \hat{y})^2,$$

with $y = 3$ and $\hat{y} = 2.5$.
Tasks:

1. Derive $\frac{\partial L}{\partial \hat{y}}$.
2. Evaluate it at $y = 3$, $\hat{y} = 2.5$.

4 Backprop Through a Single Neuron

Exercise 5: Backpropagate to the Weight

We have:

$$z = wx + b, \qquad \hat{y} = \sigma(z), \qquad L = \frac{1}{2}(y - \hat{y})^2,$$

where σ is the sigmoid.

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

Given:

$$x = 2, \quad w = 1, \quad b = -1, \quad y = 0.$$

Tasks:

1. Compute the forward pass: z, $\hat{y}$, L.
2. Compute $\frac{\partial L}{\partial \hat{y}}, \frac{\partial \hat{y}}{\partial z}, \frac{\partial z}{\partial w}, \frac{\partial L}{\partial w}$.

5 Weight Update

Exercise 6: Gradient Descent Update

Using the gradient from Exercise 5, update the weight:

$$w_{\text{new}} = w - \eta \frac{\partial L}{\partial w}.$$

Given:

$$w = 1, \quad \eta = 0.1, \quad \frac{\partial L}{\partial w} \approx 0.288.$$

Task: Compute w_{new}.

6 Two-Layer Network Forward Pass

Exercise 7: Two-Layer Network

Network:

$$h = \text{ReLU}(W_1 x + b_1), \qquad \hat{y} = W_2 h + b_2.$$

Given:

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad W_1 = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}, \quad b_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$
$$W_2 = [2 \ \ -1], \quad b_2 = 0.5.$$

Tasks:

1. Compute $z_1 = W_1 x + b_1$.
2. Apply ReLU: $h = \text{ReLU}(z_1)$.
3. Compute $\hat{y} = W_2 h + b_2$.

7 Softmax and Cross-Entropy

Exercise 8: Softmax by Hand

Given logits:

$$z = [2, 1, 0].$$

Softmax:

$$\sigma(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}.$$

Tasks:

1. Compute e^2, e^1, e^0 (symbolically and approximately).
2. Compute the denominator $\sum_j e^{z_j}$.
3. Compute the softmax probabilities $p_i = \sigma(z_i)$

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Exercise 9: Cross-Entropy Loss

Assume the true class is the second class (index 1), and

$$p = [p_0, p_1, p_2].$$

are the softmax probabilities from Exercise 8 Cross-entropy loss:

$$L = -\log(p_1).$$

Task: Compute L numerically.

Exercise 10: Gradient of Softmax + CE

Given:

$$y = [0, 1, 0]. \text{ (True one-hot label)}$$
$$p = [0.7, 0.2, 0.1].\text{(softmax output)}$$

Cross entropy:

$$L = -\sum_i y_i \log(p_i)$$

Task: Compute

$$\frac{\partial L}{\partial z_i}$$

for each i, knowing that for softmax+CE:

$$\frac{\partial L}{\partial z_i} = p_i - y_i$$

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8 Advanced: Multi-Layer Backpropagation

Exercise 11: Two-Layer Backprop (Vector Form)

Consider:

$$h = \sigma(W_1 x + b_1), \qquad \hat{y} = W_2 h + b_2,$$

with scalar loss:

$$L = \frac{1}{2}(y - \hat{y})^2.$$

here : $x \in \mathbb{R}^n, \quad W_1 \in \mathbb{R}^{m*n}, \quad b_1 \in \mathbb{R}^m, \quad W_2 \in \mathbb{R}^{1*m}, \quad b_2 \in \mathbb{R}, \quad \sigma$ applied element wise

Task: Derive expressions for:

$$\frac{\partial L}{\partial W_2}, \qquad \frac{\partial L}{\partial b_2}, \qquad \frac{\partial L}{\partial W_1}, \qquad \frac{\partial L}{\partial b_1},$$

in terms of $x, h, W_1, W_2, y,$ and $\hat{y}$.

9 Advanced: Convolution Math

Exercise 12: 1D Convolution Forward Pass

Input (length 4):

$$x = [1, 2, 0, -1]$$

Kernel (length 3):

$$w = [2, -1, 1].$$

No padding, stride 1.
Task: Compute the 1D convolution output y.

Exercise 13: Gradient of 1D Convolution w.r.t Kernel

Same setup as Exercise 12, but suppose we have a loss:

$$L = \frac{1}{2}(y_1 - t)^2,$$

where y_1 is the second output and t is some target.
Given:

$$y_1 = 3, \quad t = 1, \quad x_{\text{window}} = [2, 0, -1].$$

Task: Derive $\frac{\partial L}{\partial w_i}$ for $i = 0, 1, 2$.

10 Advanced: Vanishing Gradient Illustration

Exercise 14: Sigmoid Chain Multiplication

Consider a deep network with L layers, each using a sigmoid activation with derivative at most 0.25 in magnitude

$$|\sigma'(z)| \leq 0.25.$$

Suppose the gradient backpropagated through each layer is multiplied by a factor at most 0.25 at each step.
Task: Show how the gradient magnitude behaves as L grows (i.e. explain vanishing gradients)